

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU, Belagavi)



PROJECT REPORT

On

ProactiveHeal with Veritas:

**An AI-Driven System for Safe, Predictive Self-Healing in Cloud
Infrastructure**

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INTRODUCTION

1. Introduction

Cloud computing has become an essential part of modern digital infrastructure, supporting critical services in sectors such as healthcare, finance, education, e-commerce, and telecommunications. Organizations rely heavily on cloud platforms to ensure continuous availability, scalability, and performance of their applications. As cloud environments grow in complexity, managing infrastructure reliability and preventing service disruptions have become significant challenges.

Traditional cloud monitoring systems primarily operate in a reactive manner, identifying issues only after they occur. Failures such as high CPU utilization, excessive memory consumption, disk saturation, and network congestion can lead to degraded performance, service outages, and increased operational costs. Although existing monitoring tools can generate alerts when thresholds are exceeded, they often depend on human intervention for diagnosis and remediation, resulting in delayed recovery and increased downtime.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the development of predictive monitoring systems capable of analyzing infrastructure metrics and forecasting potential failures before they impact users. These proactive approaches provide organizations with valuable time to take corrective actions and maintain service continuity. However, a major limitation of current autonomous remediation systems is the absence of an independent mechanism to verify whether the recommended corrective action is safe, compliant, and suitable for execution.

To address this challenge, this project proposes **ProactiveHeal with Veritas**, an AI-driven cloud self-healing framework that combines predictive failure detection with verified autonomous remediation. The system continuously monitors key infrastructure metrics, including CPU utilization, memory usage, disk usage, and network utilization. Using machine learning models, it predicts potential failures before they occur and generates intelligent recommendations along with explanations and root-cause analysis.

The proposed framework introduces **Veritas**, a dedicated verification layer that acts as a safety and policy enforcement mechanism. Before any corrective action is executed, Veritas evaluates the recommendation against predefined operational rules and safety guidelines. If the action satisfies all verification requirements, it is automatically executed; otherwise, the system escalates the issue for manual review by engineers. Additionally, real-time alerts containing issue details, AI reasoning, and recommended actions are delivered to engineers through mobile notifications, enabling faster and more informed decision-making.

By integrating predictive analytics, AI-based reasoning, automated alerting, and verified remediation, ProactiveHeal with Veritas aims to create a reliable, transparent, and trustworthy self-healing cloud management system that minimizes downtime while ensuring operational safety.

PROBLEM DEFINITION

2.1 Problem Statement

Modern cloud infrastructures support a wide range of critical applications that require high availability, reliability, and performance. However, most existing cloud monitoring and management systems are reactive in nature, meaning they identify and respond to issues only after they have already affected system performance. Problems such as excessive CPU utilization, memory exhaustion, disk saturation, and network congestion can lead to service degradation, application downtime, and poor user experience.

Although current monitoring solutions generate alerts when abnormal conditions are detected, the process of diagnosing the root cause and determining appropriate corrective actions often relies heavily on human intervention. This increases the Mean Time to Repair (MTTR) and places a significant burden on cloud engineers and site reliability teams.

Recent AI-based approaches have introduced automated remediation capabilities, allowing systems to take corrective actions without human involvement. However, these systems face a critical challenge: they execute recommended actions without independently verifying whether those actions are safe, compliant with operational policies, or suitable for the current system state. An incorrect automated action may cause service interruptions, configuration errors, or even larger failures than the original problem.

Therefore, there is a need for a proactive and trustworthy cloud management framework that can:

- Predict infrastructure issues before they occur.
- Provide AI-driven reasoning and root-cause analysis.
- Alert engineers with actionable insights.
- Verify recommended remediation actions before execution.
- Ensure safe and reliable autonomous operation.

This project addresses these challenges through **ProactiveHeal with Veritas**, an AI-driven predictive self-healing framework that combines proactive failure detection with an independent verification layer to ensure that only safe and approved actions are executed.

2.2 Background Information (Literature Review)

Cloud computing has become the foundation of modern digital services, making infrastructure reliability and availability critical requirements for organizations. Traditional cloud monitoring systems primarily rely on threshold-based alerts and reactive incident management, where issues are detected only after performance degradation or service disruption has occurred. This approach often results in increased downtime, higher operational costs, and greater dependency on manual intervention by cloud engineers.

Several studies have explored the application of Artificial Intelligence for IT Operations (AIOps) to improve cloud infrastructure management. Machine learning techniques such as anomaly detection, predictive analytics, and time-series forecasting have demonstrated the ability to identify abnormal patterns in metrics such as CPU utilization, memory consumption, disk usage, and network traffic before failures occur. These approaches enable proactive maintenance and reduce the likelihood of unexpected service outages.

Recent research has also investigated self-healing cloud systems that automatically perform remediation actions, such as resource scaling, service restarting, or workload redistribution. While these systems improve response times, they often execute corrective actions without independently validating whether the actions are safe, compliant, or appropriate for the current operational environment. This can potentially lead to unintended consequences and larger system failures.

Explainable AI (XAI) has emerged as an important area of research to address the lack of transparency in AI-driven decision-making. By providing reasoning, root-cause analysis, and interpretable recommendations, explainable systems help engineers understand and trust automated decisions. Additionally, intelligent alerting mechanisms have been developed to deliver real-time notifications and actionable insights directly to operational teams.

Despite these advancements, very few solutions integrate predictive failure detection, explainable AI, automated alerting, and independent verification of remediation actions

within a single framework. The absence of a dedicated verification mechanism remains a significant limitation in existing autonomous cloud management systems. This gap motivates the development of **ProactiveHeal with Veritas**, which combines proactive anomaly prediction, AI-driven reasoning, engineer notifications, and a verification layer that validates remediation actions before execution, ensuring both operational efficiency and system safety.

OBJECTIVE

3.1 Primary Objectives

- To develop an AI-driven proactive monitoring system capable of continuously analyzing cloud infrastructure metrics such as CPU utilization, memory usage, disk usage, and network utilization.
- To predict potential system failures and performance degradation before they impact cloud services using machine learning techniques.
- To provide intelligent root-cause analysis and explainable AI-based reasoning for detected anomalies and predicted failures.
- To design and implement **Veritas**, an independent verification layer that validates proposed remediation actions against predefined operational policies and safety rules.
- To enable safe automated remediation by executing corrective actions only after successful verification through the Veritas layer.
- To generate real-time alerts and notifications for cloud engineers, providing detailed information about predicted issues, causes, and recommended actions.

3.2 Secondary Objectives

- To reduce system downtime and improve overall cloud infrastructure reliability through proactive issue detection and response.
- To minimize manual monitoring efforts and reduce the workload of cloud operations teams.
- To improve transparency and trust in AI-driven decision-making through explainable recommendations and reasoning.

- To develop a dashboard for visualizing infrastructure metrics, anomaly trends, predictions, alerts, and remediation activities.
- To create a scalable and modular framework that can be extended to support additional cloud metrics, policies, and remediation strategies in the future.
- To establish a foundation for future research publications, product development, and advanced self-healing cloud management systems.

METHODOLOGY

This chapter describes the overall approach and the step-by-step procedures followed to design and implement the proposed cloud infrastructure self-healing framework. The methodology is structured to ensure proactive failure prediction, explainable decision-making, safe remediation, and efficient cloud operations. The system combines machine learning, intelligent alerting, and a verification mechanism to provide a reliable and trustworthy cloud management solution.

4.1 Approach

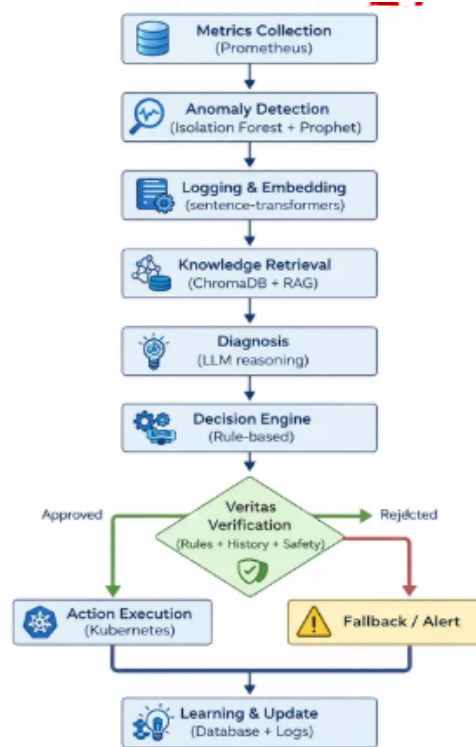


fig.illustrates the flow of proposed system

The proposed system follows a multi-stage, modular approach for proactive cloud infrastructure management. Instead of relying solely on traditional monitoring systems that react after failures occur, the framework continuously analyzes cloud resource metrics and predicts potential issues before they affect system performance. The system decomposes the problem into multiple interconnected components, each addressing a specific operational requirement such as monitoring, prediction, explainability, alert generation, verification, and remediation.

The overall approach consists of the following key stages:

1. Infrastructure Monitoring

The system continuously monitors critical cloud infrastructure metrics including CPU utilization, memory usage, disk usage, and network utilization. These metrics provide a comprehensive view of system health and serve as the primary input for predictive analysis.

2. Failure Prediction using Machine Learning

A machine learning model analyzes historical and real-time metric data to identify abnormal trends and predict potential failures before they occur. This proactive

approach enables the system to take preventive measures rather than reacting to outages after they have already impacted users.

3. **AI-Based Root Cause Analysis**

When a potential issue is detected, the AI engine analyzes the contributing factors and generates an explanation describing the likely cause of the problem. This allows engineers to understand why the issue is expected to occur and increases trust in the system's recommendations.

4. **Alert Generation and Notification**

Based on the predicted issue, the system generates alerts containing the detected anomaly, probable cause, severity level, and recommended action. These alerts are delivered to engineers through mobile notifications, enabling rapid awareness and response.

5. **Remediation Recommendation**

The AI engine proposes a suitable remediation action based on the predicted failure and historical system behavior. Examples include resource scaling, service restart, workload redistribution, or other corrective measures.

6. **Veritas Verification Layer**

The proposed remediation action is passed to the Veritas layer, which acts as an independent verification mechanism. Veritas evaluates the action against predefined operational policies, safety constraints, and organizational rules to ensure that the recommended action is safe and appropriate for execution.

7. **Automated or Manual Remediation**

If Veritas approves the proposed action, the system automatically executes the remediation process. If the action violates any rule or fails verification, automatic execution is blocked and the issue is escalated for manual intervention by cloud engineers.

8. **System Monitoring and Feedback**

After remediation, the system continues monitoring infrastructure metrics to verify recovery and maintain operational stability. The collected information can be utilized to improve future predictions and decision-making processes.

Flow of the Proposed System

The methodological flow of the system can be summarized as follows:

1. Cloud infrastructure metrics collection (CPU, Memory, Disk, Network)
2. Machine learning–based analysis of infrastructure data
3. Prediction of potential failures and anomalies
4. AI-based root cause analysis and reasoning generation
5. Alert generation and notification to engineers
6. Remediation action recommendation
7. Verification through Veritas layer
8. Automatic execution if approved
9. Manual intervention if rejected
10. Continuous monitoring and feedback collection

This modular design ensures interpretability, flexibility, and ease of future extension.

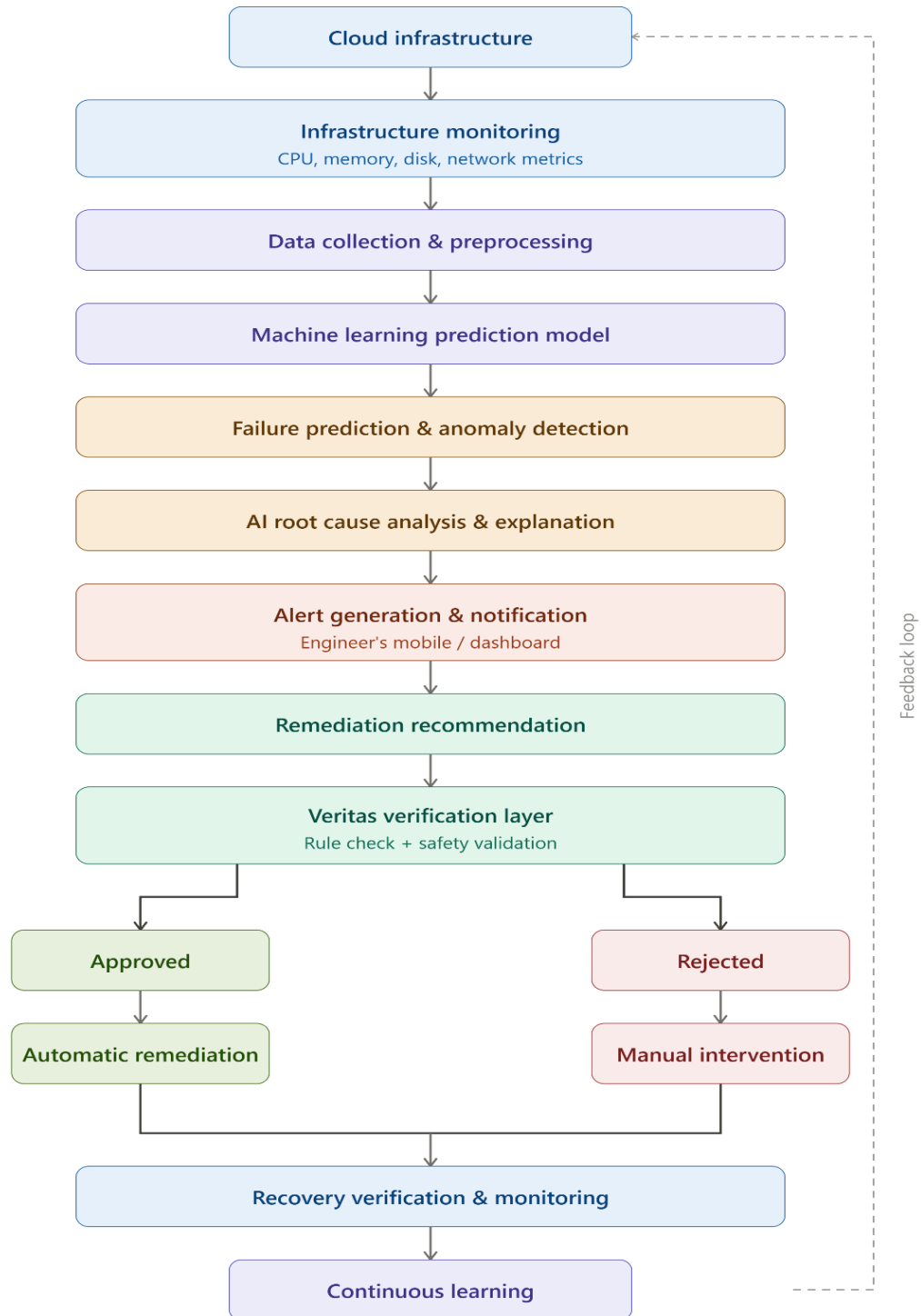


Figure 4.1.illustrates the overall workflow of the proposed ProactiveHeal with Veritas framework. The system continuously monitors cloud infrastructure metrics, predicts failures using machine learning, performs AI-based root cause analysis, generates alerts for engineers, and recommends remediation actions.

4.2 Procedures

The execution of the project follows a structured and sequential procedure to ensure systematic development and evaluation of the proposed ProactiveHeal with Veritas framework.

Infrastructure Monitoring Setup

The first stage involves configuring the monitoring environment to continuously collect cloud infrastructure metrics such as CPU utilization, memory usage, disk usage, and network utilization. These metrics serve as the primary source of information for system analysis and prediction.

Data Collection and Preprocessing

The collected metrics are organized, cleaned, and prepared for analysis. Historical and real-time data are processed to identify resource utilization patterns and establish baseline system behavior.

Machine Learning Model Development

A machine learning model is developed and trained using infrastructure metric data. The model learns normal system behavior and identifies patterns that may indicate potential failures or abnormal operating conditions.

Failure Prediction and Anomaly Detection

The trained model is deployed to continuously analyze incoming metrics and predict potential failures before they affect system performance. Detected anomalies are classified based on their severity and potential impact.

AI-Based Reasoning and Root Cause Analysis

When a potential issue is identified, the AI engine analyzes the contributing factors and generates explanations describing the probable cause of the anomaly. This provides engineers with meaningful insights into the predicted issue.

Alert Generation and Notification

The system generates alerts containing the predicted issue, severity level, root cause analysis, and recommended corrective action. These alerts are delivered to engineers through mobile notifications and monitoring dashboards.

Remediation Recommendation Generation

Based on the predicted failure and historical system behavior, the AI module recommends suitable remediation actions to prevent service disruption.

Veritas Verification Process

The proposed remediation action is passed to the Veritas layer, where it is evaluated against predefined operational policies, safety constraints, and organizational rules. This verification process ensures that only safe and compliant actions are approved.

Automated or Manual Remediation

If the proposed action is approved by Veritas, the system automatically executes the remediation process. If the action fails verification, automatic execution is blocked and the issue is escalated to engineers for manual intervention.

System Integration and Evaluation

All modules are integrated into a unified framework and tested under various simulated cloud scenarios to evaluate prediction accuracy, alert effectiveness, verification reliability, and overall system performance.

Timelines and Milestones

- Infrastructure monitoring setup and metric collection
- Data preprocessing and baseline analysis
- Machine learning model development and training
- Failure prediction and anomaly detection integration
- AI reasoning and root cause analysis implementation
- Alert generation and notification system development
- Veritas verification layer implementation
- Automated remediation workflow development
- System integration and testing
- Performance evaluation and result analysis
- Documentation, report preparation, and final presentation

Each milestone is completed sequentially to ensure correctness, reliability, and successful integration before proceeding to the next phase of development.

PROJECT EXECUTION

This chapter describes how the proposed ProactiveHeal with Veritas framework was planned, designed, and implemented. The focus is on transforming the conceptual architecture into a functional cloud infrastructure management system while addressing practical challenges related to failure prediction, automated remediation, operational safety, and explainability.

5.1 Planning and Design

The initial planning phase involved defining the scope of the project and identifying a feasible approach to address the limitations of traditional cloud monitoring systems. Early brainstorming sessions focused on understanding common cloud infrastructure failures, existing monitoring solutions, and the challenges associated with autonomous remediation in production environments.

The original design objective was to develop a proactive cloud management system capable of predicting failures before they occur and automatically resolving them. However, it was recognized that fully autonomous remediation without validation could introduce significant operational risks. As a result, **the design was refined to incorporate an independent verification mechanism capable of evaluating proposed actions before execution.**

Key design decisions made during this phase included:

- Using infrastructure metrics such as CPU utilization, memory usage, disk usage, and network utilization as the primary indicators of system health.
- Adopting a machine learning–based approach for failure prediction instead of relying solely on static threshold-based monitoring.

- Incorporating AI-driven reasoning and root-cause analysis as a core system component to improve transparency and explainability.
- Implementing a real-time alerting mechanism to notify engineers about predicted failures and recommended actions.
- Introducing the Veritas verification layer as the primary novel contribution of the project to ensure safe and policy-compliant remediation.
- Allowing automatic execution only for actions approved by Veritas while escalating rejected actions for manual intervention.
- Designing a lightweight dashboard for visualizing infrastructure metrics, predictions, alerts, and remediation activities.

High-level system architecture diagrams and workflow designs were created to visualize the interaction between monitoring, prediction, reasoning, alert generation, verification, and remediation modules. This planning phase ensured that each component served a clearly defined purpose and could be independently developed, tested, and integrated into the final framework.

5.2 Implementation

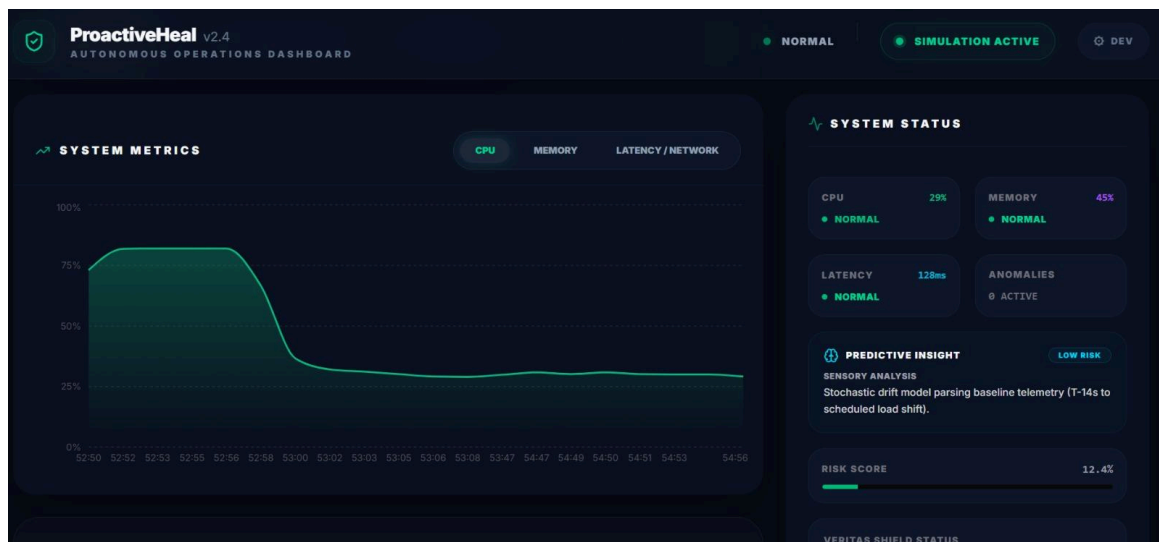


fig. demonstrates the baseline normal states

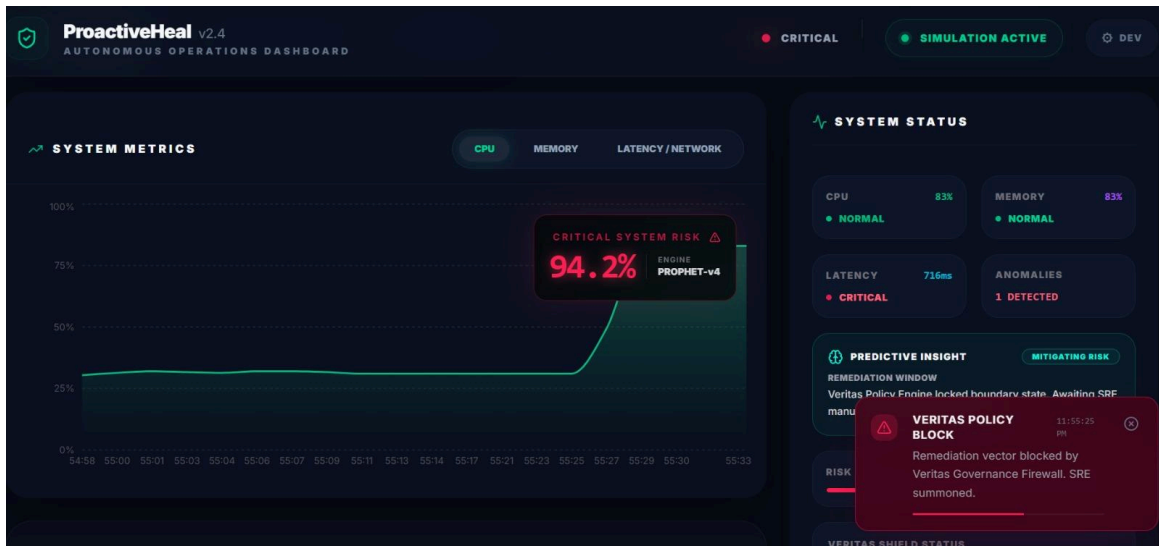


fig. demonstrates anomaly and implementation of the pipeline

The implementation phase translated the planned design into a functional cloud self-healing framework through a series of well-defined development stages.

The first stage involved implementing the infrastructure monitoring module responsible for continuously collecting cloud metrics such as CPU utilization, memory usage, disk usage, and network utilization. These metrics were organized and processed to provide a consistent stream of operational data for analysis.

Once the monitoring pipeline was established, the machine learning prediction module was developed to analyze historical and real-time infrastructure metrics. The model was designed to identify abnormal patterns and forecast potential failures before they could affect service availability and system performance.

Following the prediction stage, an AI reasoning module was implemented to analyze detected anomalies and generate root-cause explanations. This component provides engineers with meaningful insights regarding the factors contributing to a predicted failure, improving trust and interpretability of the system's recommendations.

The alert generation module was then developed to create detailed notifications containing information about predicted issues, severity levels, probable causes, and suggested

remediation actions. These alerts are delivered to engineers through mobile devices and monitoring dashboards to ensure rapid awareness and response.

The next stage focused on implementing the Veritas verification layer. This module evaluates every recommended remediation action against predefined operational rules, safety policies, and organizational constraints. The verification process determines whether the proposed action is safe for automatic execution or requires human review.

Once verification functionality was established, the remediation workflow was developed. Approved actions are executed automatically to prevent service disruption, while rejected actions are escalated to engineers for manual intervention. This approach combines the efficiency of automation with the safety of human oversight.

Finally, all system components were integrated into a unified framework. The integrated prototype enables continuous monitoring, proactive failure prediction, AI-based reasoning, intelligent alerting, verified remediation, and recovery verification within a single operational workflow. This integration resulted in a cohesive and practical cloud infrastructure management system capable of reducing downtime while maintaining operational safety.

TOOLS AND TECHNIQUES USED

This chapter describes the software tools, libraries, and techniques used in the development of the proposed ProactiveHeal with Veritas framework. Each tool and method was selected to ensure efficient monitoring, accurate failure prediction, explainable decision-making, safe remediation, and seamless system integration within a cloud infrastructure environment.

6.1 Tools

Programming and Development Tools :

Python

Python was used as the primary programming language for the entire project. It was utilized for data processing, machine learning model development, anomaly detection, AI reasoning, alert generation, verification logic, and overall system integration. Python was selected due to its simplicity, extensive library support, and suitability for cloud and AI-based applications.

Visual Studio Code (VS Code)

Visual Studio Code was used as the integrated development environment for writing, debugging, and managing the project source code. It facilitated efficient development and testing of various modules within the framework.

Monitoring and Visualization Tools

Prometheus

Prometheus was used to collect and store infrastructure metrics such as CPU utilization, memory usage, disk usage, and network utilization. These metrics serve as the primary input for anomaly detection and failure prediction.

Machine Learning and Prediction Tools:

Isolation Forest

Isolation Forest was used as the anomaly detection algorithm for identifying abnormal patterns in cloud infrastructure metrics. It enables early detection of unusual system behavior without requiring labeled failure data.

Prophet

Prophet was used for time-series forecasting of infrastructure metrics. It analyzes historical trends and predicts future resource utilization patterns, enabling proactive identification of potential failures.

Knowledge Management and AI Tools :

Sentence Transformers

Sentence Transformer models were used to convert incident descriptions and system events into vector representations. These embeddings enable semantic comparison between current and historical incidents.

ChromaDB

ChromaDB was used as the vector database for storing historical incidents and their corresponding remediation actions. It supports semantic retrieval of similar incidents to improve decision-making and reasoning.

AI Reasoning Engine

An AI reasoning module was implemented to analyze detected anomalies, perform root-cause analysis, and generate explanations and remediation recommendations for engineers.

Verification and Management Tools:

Veritas Verification Layer

Veritas is the custom verification module developed as the primary contribution of the project. It evaluates recommended remediation actions against predefined operational policies, safety constraints, and organizational rules before execution.

Cloud Management Tools :

Kubernetes (Simulated Environment)

Kubernetes was used to simulate cloud infrastructure operations and remediation activities. It provides a realistic environment for testing scaling, service restart, and workload management actions.

Alerting and Communication Tools:

Alert Notification System

A notification system was developed to deliver real-time alerts to engineers whenever anomalies or potential failures are detected. The alerts include issue details, AI-generated reasoning, severity levels, and recommended actions.

6.2 Techniques

Anomaly Detection

Anomaly detection techniques were used to identify unusual patterns in cloud infrastructure metrics such as CPU utilization, memory usage, disk usage, and network utilization. By continuously analyzing metric behavior, the system can detect abnormalities that may indicate potential failures before they affect service performance.

Time-Series Forecasting

Time-series forecasting was applied to predict future trends in infrastructure metrics using historical monitoring data. This technique enables the system to identify potential resource bottlenecks and performance degradation before they occur, allowing proactive intervention.

Predictive Analytics

Predictive analytics techniques were incorporated to forecast potential system failures based on historical and real-time infrastructure data. Instead of reacting to incidents after they occur, the framework predicts failures in advance and provides sufficient time for corrective actions.

AI-Based Root Cause Analysis

Artificial Intelligence techniques were utilized to analyze detected anomalies and determine the most probable cause of a predicted failure. This helps engineers understand the underlying reason behind system behavior and supports informed decision-making.

Explainable Artificial Intelligence (XAI)

Explainable AI concepts were integrated into the framework to provide understandable reasoning behind predictions and recommended remediation actions. This improves transparency, trust, and usability by allowing engineers to understand why a particular issue was detected and why a specific action was recommended.

Semantic Similarity Retrieval

Semantic retrieval techniques were employed using vector embeddings and similarity search. Historical incidents stored in the knowledge base are compared with current anomalies to identify similar past events and retrieve relevant remediation experiences.

Rule-Based Verification

A rule-based verification approach was implemented through the Veritas layer. Before any remediation action is executed, it is evaluated against predefined operational policies, safety rules, confidence thresholds, and organizational constraints. This ensures that only safe and approved actions are executed automatically.

Automated Remediation

Automated remediation techniques were used to perform corrective actions when approved by the Veritas verification layer. Examples include service restart, resource scaling, workload redistribution, and other predefined recovery procedures. This significantly reduces response time and system downtime.

Alerting and Notification Mechanism

An intelligent alerting mechanism was developed to notify engineers whenever anomalies or predicted failures are detected. Alerts include issue details, severity level, root-cause analysis, AI reasoning, and recommended actions, enabling rapid response and decision-making.

Human-in-the-Loop Decision Making

For remediation actions that fail verification or require additional approval, a human-in-the-loop approach is adopted. Engineers receive notifications and manually evaluate the situation before executing corrective actions, ensuring operational safety.

Modular System Architecture

The framework follows a modular architecture where monitoring, prediction, reasoning, alerting, verification, and remediation are implemented as independent but interconnected components. This design improves scalability, maintainability, and flexibility.

RESULTS AND DISCUSSION

This chapter presents the final results obtained from the proposed deep learning–based osteoarthritis assessment system and discusses their significance with respect to the stated objectives. The evaluation focuses on overall system functionality, interpretability, and practical applicability rather than clinical deployment performance.

7.1 Final Results

This chapter presents the final results obtained from the proposed ProactiveHeal with Veritas framework and discusses their significance with respect to the stated objectives.

For a given cloud infrastructure environment, the system produces the following outputs:

- Continuous monitoring of CPU utilization, memory usage, disk usage, and network utilization
- Detection of abnormal system behavior and anomalies
- Prediction of potential failures before service degradation occurs
- AI-generated root cause analysis and reasoning
- Real-time alerts and notifications for engineers
- Recommended remediation actions
- Veritas verification decision (Approved / Rejected)
- Automatic remediation for approved actions
- Manual intervention requests for rejected actions
- Dashboard visualization of metrics, predictions, alerts, and remediation activities

The infrastructure monitoring module successfully collected and processed cloud resource metrics, providing a reliable foundation for anomaly detection and predictive analysis. The machine learning prediction model demonstrated the ability to identify abnormal trends and forecast potential failures before they impacted system performance.

The AI reasoning module generated meaningful explanations describing the probable causes of detected anomalies. These explanations improved transparency and enabled engineers to better understand the reasoning behind predicted failures and recommended corrective actions.

The alert generation system successfully delivered notifications containing issue details, severity levels, probable causes, and recommended actions. This ensured that engineers remained informed about potential incidents and could respond quickly when required.

The Veritas verification layer effectively evaluated remediation actions against predefined operational policies and safety constraints. Approved actions were executed automatically, while unsafe or non-compliant actions were blocked and escalated for manual review. This demonstrated the feasibility of combining autonomous operation with safety-aware decision making.

The dashboard interface successfully displayed infrastructure metrics, anomaly trends, prediction results, alerts, and verification outcomes through a centralized monitoring platform. This provided clear visibility into system status and decision-making processes.

7.2 Discussion

The results indicate that the proposed framework fulfills its primary objectives of proactive failure prediction, explainable decision-making, intelligent alerting, and verification-based autonomous remediation within a single integrated system. The modular architecture allows each component to operate independently while contributing to overall system reliability and operational safety.

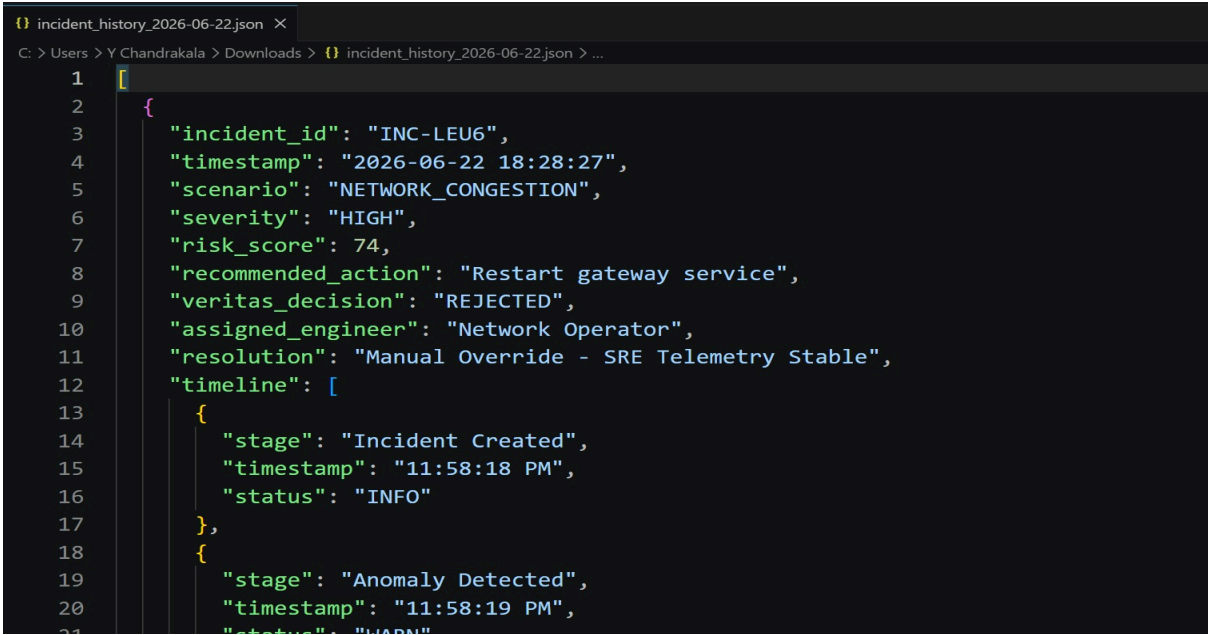
The inclusion of machine learning-based prediction enables the framework to identify potential failures before they result in service disruption. This proactive approach represents a significant improvement over traditional reactive monitoring systems that respond only after failures have occurred.

The AI reasoning component enhances interpretability by providing root-cause explanations for detected anomalies. This transparency improves engineer trust in the system and supports informed operational decision-making.

The Veritas verification layer serves as the primary innovation of the project. Unlike conventional self-healing systems that directly execute recommended actions, Veritas introduces an additional safety and policy validation stage. This reduces the risk of unsafe autonomous actions and ensures that remediation decisions remain aligned with organizational requirements.

The integration of alert notifications and human-in-the-loop decision making further strengthens operational reliability. Engineers remain informed throughout the process and can intervene whenever automated remediation is considered unsafe or inappropriate.

Overall, the proposed ProactiveHeal with Veritas framework demonstrates a feasible, explainable, and safety-aware approach to cloud infrastructure management. By combining predictive analytics, AI reasoning, intelligent alerting, and verification-based remediation, the system provides a practical foundation for the next generation of trustworthy self-healing cloud environments.



```
1  [
2  {
3    "incident_id": "INC-LEU6",
4    "timestamp": "2026-06-22 18:28:27",
5    "scenario": "NETWORK_CONGESTION",
6    "severity": "HIGH",
7    "risk_score": 74,
8    "recommended_action": "Restart gateway service",
9    "veritas_decision": "REJECTED",
10   "assigned_engineer": "Network Operator",
11   "resolution": "Manual Override - SRE Telemetry Stable",
12   "timeline": [
13     {
14       "stage": "Incident Created",
15       "timestamp": "11:58:18 PM",
16       "status": "INFO"
17     },
18     {
19       "stage": "Anomaly Detected",
20       "timestamp": "11:58:19 PM",
21       "status": "WARN"
```

Figure 7.1: Data being stored in the format of a json file, which includes: incident id, timestamp, scenario, severity, etc.

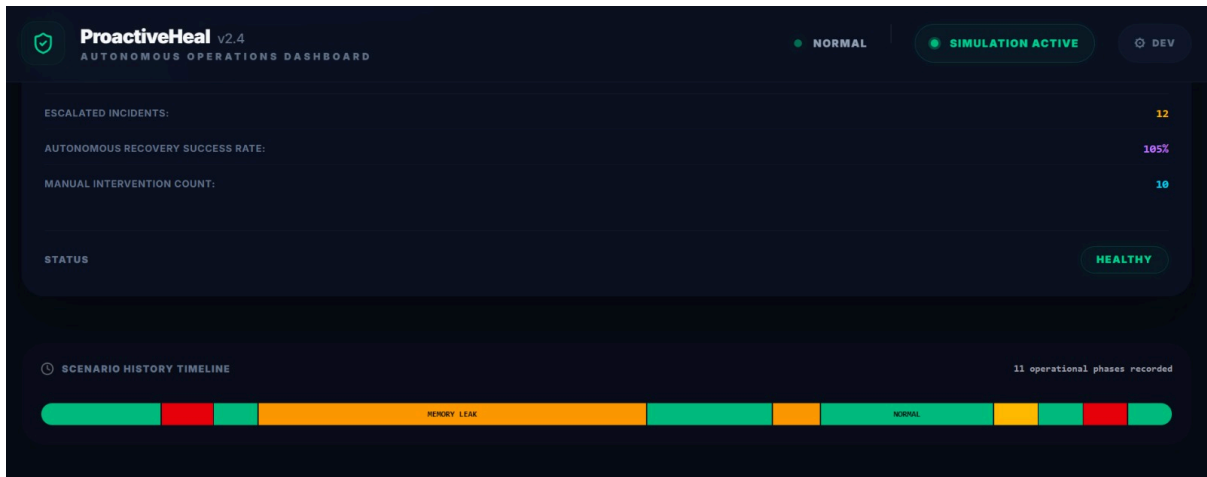


Figure 7.2: Scenario History Timeline

PROTOTYPE

This chapter describes the developed prototype of the proposed ProactiveHeal with Veritas framework. The prototype is implemented as a software-based cloud infrastructure management and decision-support system, demonstrating the feasibility of integrating predictive analytics, AI reasoning, alert generation, verification-based remediation, and infrastructure monitoring into a single interactive platform.

8.1 Prototype Description

The prototype is a software-based system designed to proactively monitor cloud infrastructure, predict potential failures, provide AI-generated explanations, and execute safe remediation actions through an independent verification mechanism. It functions as a proof-of-concept self-healing cloud management framework rather than a production-ready deployment platform.

The prototype allows users to:

- Monitor cloud infrastructure metrics such as CPU utilization, memory usage, disk usage, and network utilization.
- Detects anomalies and abnormal resource utilization patterns.
- Predict potential infrastructure failures before service degradation occurs.

- View AI-generated root-cause analysis and reasoning for detected issues.
- Receive real-time alerts and notifications regarding predicted failures.
- Obtain recommended remediation actions based on detected anomalies.
- View Veritas verification decisions (Approved / Rejected).
- Execute approved remediation actions automatically.

The system integrates monitoring, prediction, reasoning, verification, alerting, and remediation modules into a unified workflow.

8.2 Development Process

The prototype development followed a modular and incremental approach to ensure stability, correctness, and successful integration of all system components.

Initially, the infrastructure monitoring module was developed to collect CPU utilization, memory usage, disk usage, and network utilization metrics from the cloud environment. The collected metrics were organized and prepared for analysis.

Once the monitoring pipeline was established, the machine learning prediction module was implemented to analyze historical and real-time infrastructure data. The model was trained to identify abnormal patterns and forecast potential failures before they could affect service performance.

Following successful prediction testing, the AI reasoning module was developed to perform root-cause analysis and generate explanations describing the probable causes of detected anomalies. This module improves transparency and assists engineers in understanding system behavior.

The alert generation system was then integrated to provide notifications containing issue details, severity levels, predicted impacts, root-cause information, and recommended remediation actions.

The Veritas verification layer was subsequently implemented as the primary novel component of the framework. This module evaluates all recommended remediation actions against predefined operational rules, safety policies, and organizational constraints before execution.

After verification functionality was established, the automated remediation workflow was developed. Approved actions are executed automatically, while rejected actions are forwarded to engineers for manual review and intervention.

Finally, all modules were integrated into a unified dashboard-driven platform. The interface was refined iteratively to provide clear visualization of infrastructure metrics, predictions, alerts, verification outcomes, and remediation activities, resulting in a cohesive cloud management prototype.

8.3 Testing and Validation

The prototype was tested using simulated cloud infrastructure scenarios to evaluate functional correctness, module integration, and overall system reliability.

Testing focused on the following aspects:

- Correct collection and processing of CPU, memory, disk, and network utilization metrics.
- Successful detection of anomalies and abnormal infrastructure behavior.
- Accurate prediction of potential failures using machine learning models.
- Proper generation of AI-based root-cause analysis and reasoning.
- Successful delivery of alerts and notifications to engineers.
- Correct recommendation of remediation actions based on detected anomalies.
- Proper verification of remediation actions through the Veritas layer.
- Successful execution of approved remediation actions.
- Correct escalation of rejected actions for manual intervention.
- Accurate visualization of metrics, alerts, predictions, and verification results through the dashboard.

Functional testing confirmed that all modules operated correctly and interacted successfully within the integrated framework. The system consistently generated predictions, explanations, alerts, verification decisions, and remediation recommendations without significant operational issues.

It is important to note that this testing validates the functionality and integration of the proposed framework rather than large-scale production deployment performance. The prototype is intended as a proof-of-concept demonstration of proactive, explainable, and safety-aware cloud infrastructure management.

CONCLUSION

9.1 Summary

This project presented the design and implementation of **ProactiveHeal with Veritas**, an AI-driven framework for proactive cloud infrastructure management and verified self-healing. The system addresses key limitations of conventional cloud monitoring approaches by predicting potential failures before they occur while ensuring that remediation actions are executed only after undergoing safety and policy verification.

The proposed framework integrates multiple components, including infrastructure monitoring, machine learning-based failure prediction, AI-driven root-cause analysis,

intelligent alert generation, remediation recommendation, and the novel Veritas verification layer. The system continuously monitors critical cloud metrics such as CPU utilization, memory usage, disk usage, and network utilization to identify abnormal patterns and forecast potential failures.

A key contribution of the project is the introduction of the Veritas verification layer, which evaluates recommended remediation actions against predefined operational rules and safety policies before execution. This approach reduces the risks associated with autonomous decision-making and provides a balance between automation and operational safety. In addition, the framework delivers real-time alerts and AI-generated explanations to engineers, improving transparency and supporting informed decision-making.

The developed prototype successfully demonstrates end-to-end functionality, providing continuous monitoring, anomaly detection, failure prediction, root-cause analysis, intelligent notifications, verified remediation, and recovery verification within a unified platform. By emphasizing explainability, safety, and proactive management, the project demonstrates a practical approach to improving cloud infrastructure reliability and reducing service downtime.

Overall, the project achieves its stated objectives and illustrates how artificial intelligence, predictive analytics, and verification mechanisms can be combined to create a trustworthy and efficient self-healing cloud management framework.

9.2 Personal Reflection

This project provided valuable hands-on experience in applying artificial intelligence and machine learning concepts to real-world cloud infrastructure management challenges. It strengthened understanding of predictive analytics, anomaly detection, cloud monitoring, and intelligent automation while highlighting the importance of safety and explainability in autonomous systems.

Working on this project reinforced the significance of designing reliable systems that balance automation with human oversight. The development of the Veritas verification layer

demonstrated the importance of validating AI-generated recommendations before execution, ensuring that automation remains safe and trustworthy.

The project also improved skills in system design, software development, debugging, integration, and problem-solving. Combining multiple modules such as monitoring, prediction, reasoning, alert generation, verification, and remediation into a unified framework provided valuable experience in building complex intelligent systems.

Additionally, developing dashboards, alerting mechanisms, and explainable AI components enhanced understanding of how technical solutions can be translated into practical tools for engineers and decision-makers. The project emphasized the importance of transparency and user trust when deploying AI-based systems in critical operational environments.

Overall, this experiential learning project contributed significantly to both technical competence and analytical thinking while fostering a deeper appreciation for responsible, explainable, and safety-aware artificial intelligence in cloud computing and infrastructure management.

VISUALS

This section presents the visual elements used in the project to illustrate the architecture, workflow, monitoring process, verification mechanism, and operational outputs of the proposed ProactiveHeal with Veritas framework. These visuals enhance understanding of the system design and demonstrate the practical functioning of the developed prototype.

The following visuals are included in the report:

- **System Architecture Diagram**

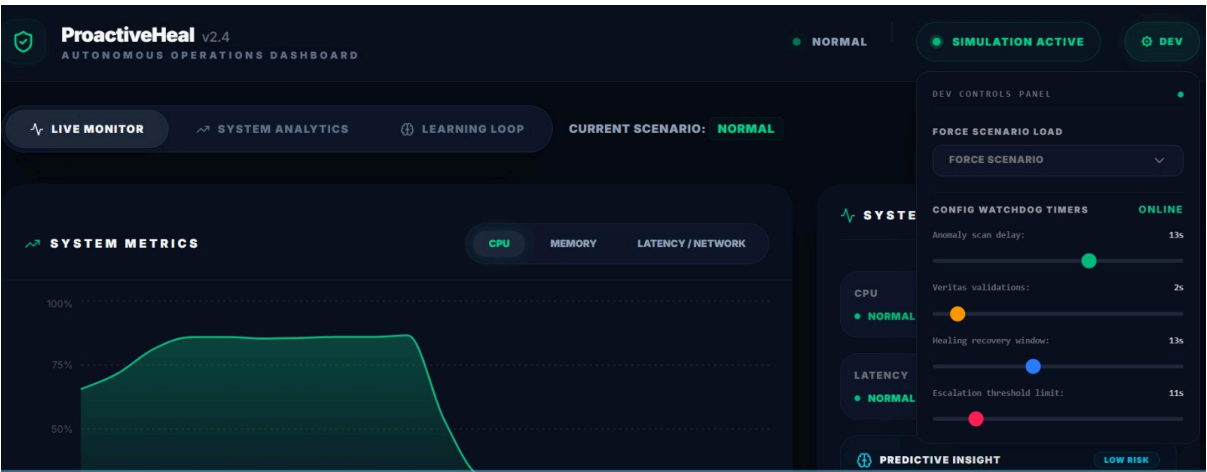


Figure 10.1: Overall architecture of the ProactiveHeal with Veritas framework showing monitoring, prediction, reasoning, verification, and remediation components.

- **Incident History and Remediation Dashboard**

The incident history dashboard maintains records of previously detected anomalies, risk scores, verification outcomes, assigned engineers, and remediation statuses. It provides traceability and auditing capabilities for infrastructure operations

A	B	C	D	E	F	G	H	I	J	K	L	M	N
incident_id	timestamp	scenario	severity	risk_score	veritas_decision	assigned_engineer	remediation_status						
INC-MOOI	#####	MEMORY	CRITICAL	89	REJECTED	Backend E	Manual Override - SRE Telemetry Stable						
INC-4WW	#####	HIGH_CPL	HIGH	85	APPROVED	Infrastruc	Automated Remediation Execution Successful						
INC-74MB	#####	NETWORK	HIGH	73	REJECTED	Network C	Manual Override - SRE Telemetry Stable						
INC-3WS2	#####	HIGH_CPL	HIGH	75	APPROVED	Infrastruc	Automated Remediation Execution Successful						
INC-MQB8	#####	NETWORK	HIGH	73	REJECTED	Network C	Manual Override - SRE Telemetry Stable						
INC-6MBH	#####	CASCADIN	CRITICAL	90	REJECTED	Lead Syste	Manual Override - SRE Telemetry Stable						
INC-9VFO	#####	MEMORY	CRITICAL	88	REJECTED	Backend E	Manual Override - SRE Telemetry Stable						
INC-P79K	#####	HIGH_CPL	HIGH	75	APPROVED	Infrastruc	Automated Remediation Execution Successful						
INC-U8HM	#####	CASCADIN	CRITICAL	94	REJECTED	Lead Syste	Manual Override - SRE Telemetry Stable						

Figure 10.2: Historical incident dashboard displaying anomaly severity, risk scores, Veritas decisions, assigned engineers, and remediation outcomes.

- **Incident History and Veritas Decision Dashboard**

The incident history dashboard provides a comprehensive record of previously detected infrastructure anomalies, associated risk scores, Veritas verification decisions, assigned engineers, and remediation outcomes. The dashboard enables auditing, operational transparency, and analysis of both automated and manually approved recovery actions.

Remix: Remix: Remix: Remix: ProactiveHeal: Self-Healing Clo... Remix La Device

ProactiveHeal v2.4
AUTONOMOUS OPERATIONS DASHBOARD

NORMAL **SIMULATION ACTIVE** DEV

INCIDENT ID	TIMESTAMP	SCENARIO	SEVERITY	RISK SCORE	VERITAS DECISION	ASSIGNED ENGINEER	REMEDIATION STATUS
INC-1112	2026-06-17 16:00:00	HIGH CPU LOAD	HIGH	87%	PASS	Infrastructure Engineer	Automated Remediation Execution Successful
INC-1111	2026-06-17 12:30:00	HIGH CPU LOAD	HIGH	75%	PASS	Infrastructure Engineer	Automated Remediation Execution Successful
INC-1110	2026-06-17 09:00:00	VERITAS 3X FAIL	CRITICAL	96%	BLOCK	P1 Security SRE Lead	Manual Override - SRE Telemetry Stable
INC-1109	2026-06-17 05:30:00	VERITAS 3X FAIL	CRITICAL	95%	BLOCK	P1 Security SRE Lead	Manual Override - SRE Telemetry Stable
INC-1108	2026-06-17 02:00:00	VERITAS 3X FAIL	CRITICAL	93%	BLOCK	P1 Security SRE Lead	Manual Override - SRE Telemetry Stable
INC-1107	2026-06-16 22:30:00	HIGH CPU LOAD	HIGH	87%	PASS	Infrastructure Engineer	Automated Remediation Execution Successful
INC-1106	2026-06-16 19:00:00	MEMORY LEAK	CRITICAL	88%	BLOCK	Backend Engineer	Manual Override - SRE Telemetry Stable
INC-1105	2026-06-16 15:30:00	HIGH CPU LOAD	HIGH	85%	PASS	Infrastructure Engineer	Automated Remediation Execution Successful
INC-1104	2026-06-16 12:00:00	CASCADING FAILURE	CRITICAL	93%	BLOCK	Lead Systems Architect	Manual Override - SRE Telemetry Stable

Figure 10.3: Incident history dashboard displaying infrastructure incidents, severity levels, risk scores, Veritas approval and rejection decisions, assigned engineers, and remediation outcomes.

OUTCOME OF WORK

The outcome of this work is the development of a functional software prototype that demonstrates a proactive, explainable, and safety-aware cloud infrastructure management framework. The proposed ProactiveHeal with Veritas system integrates machine

learning-based failure prediction, AI-driven reasoning, intelligent alerting, and verification-based remediation into a unified self-healing cloud management solution.

Outcome Summary

A complete end-to-end framework for proactive cloud infrastructure monitoring and management using CPU utilization, memory usage, disk usage, and network utilization metrics.

Integration of anomaly detection, failure prediction, AI-based root-cause analysis, intelligent alert generation, remediation recommendation, and verification-based decision making into a single system.

Development of the novel **Veritas verification layer**, which validates remediation actions against predefined operational policies and safety constraints before execution.

Implementation of automated remediation for approved actions and human-in-the-loop intervention for rejected actions, ensuring operational safety and reliability.

Development of an interactive dashboard for monitoring infrastructure metrics, visualizing anomalies, reviewing historical incidents, and tracking remediation activities.

Creation of a proof-of-concept self-healing cloud management prototype suitable for academic demonstration, research exploration, and future industrial enhancement.

Publication / Patenting / Product Development

Publication

The project is suitable for further refinement and potential publication as a research paper in areas such as cloud computing, AIOps, autonomous systems, infrastructure reliability, and explainable artificial intelligence. Additional experimentation and large-scale validation can strengthen its suitability for conference or journal submission.

Patenting

No patent has been filed for the current prototype. However, the Veritas verification layer and its integration with AI-driven self-healing cloud infrastructure present opportunities for future intellectual property development after further innovation and validation.

Product Development

The developed prototype serves as a conceptual self-healing cloud operations platform. With additional integration into real cloud environments, advanced machine learning models, larger operational datasets, and enterprise-grade deployment mechanisms, the framework can be extended into a practical cloud reliability and infrastructure automation product.

The project successfully meets the objectives of experiential learning by transforming theoretical concepts from artificial intelligence, cloud computing, machine learning, and system automation into a working prototype. It emphasizes proactive failure management, operational transparency, explainability, and safe autonomous decision-making within modern cloud infrastructure environments.

REFERENCES

[1] Z. Wang et al., “RCAgent: Cloud root cause analysis by autonomous agents with tool-augmented large language models,” arXiv preprint, 2024. Available: <https://arxiv.org/>

- [2] Y. Chen et al., “AIOpsLab: A holistic framework to evaluate AI agents for autonomous clouds,” arXiv preprint, 2025. Available: <https://arxiv.org/>
- [3] Z. Yang et al., “Intelligent fault self-healing mechanism via large language models and deep reinforcement learning,” arXiv preprint, 2025. Available: <https://arxiv.org/>
- [4] C. Ji et al., “Cloud-based AI systems using large language models for fault detection and self-healing,” arXiv preprint, 2025. Available: <https://arxiv.org/>
- [5] M. Boluda-Prieto et al., “Resilient edge-to-cloud self-healing architecture,” arXiv preprint, 2026 . Available: <https://arxiv.org/>
- [6] P. Ye et al., “NeSy-Edge: Neuro-symbolic trustworthy self-healing,” arXiv preprint, 2026. [Online]. Available: <https://arxiv.org/>
- [7] W. Xu, L. Huang, A. Fox, D. Patterson, and M. I. Jordan, “Detecting large-scale system problems by mining console logs,” in Proc. ACM SIGOPS Symp. Operating Systems Principles (SOSP), 2009, doi: 10.1145/1629575.1629587.
- [8] M. Chen, A. X. Zheng, J. Lloyd, M. I. Jordan, and E. Brewer, “Failure diagnosis using decision trees,” in Proc. Int. Conf. Autonomic Computing (ICAC), 2004, doi: 10.1109/ICAC.2004.1301347.
- [9] N. Laptev, S. Amizadeh, and I. Flint, “Generic and scalable framework for automated time-series anomaly detection,” in Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD), 2015, doi: 10.1145/2783258.2788611.
- [10] P. Bodík, G. Friedman, L. Biewald, H. Levine, G. Candea, and A. Fox, “Combining visualization and statistical analysis to improve operator confidence and efficiency for failure detection and localization,” in Proc. USENIX Symp. Networked Systems Design and Implementation (NSDI), 2010.
- [11] B. Burns, B. Grant, D. Oppenheimer, E. Brewer, and J. Wilkes, “*Borg, Omega, and Kubernetes: Lessons learned from three container-management systems over a decade,*” Commun. ACM, vol. 59, no. 5, pp. 50–57, 2016, doi: 10.1145/2890784.
- [12] T. Lorigo-Botran, J. Miguel-Alonso, and J. A. Lozano, “*A review of auto-scaling techniques for elastic applications in cloud environments,*” J. Grid Comput., vol. 12, no. 4, pp. 559–592, 2014, doi: 10.1007/s10723-014-9314-7.
- [13] J. Dean and L. A. Barroso, “*The tail at scale,*” Commun. ACM, vol. 56, no. 2, pp. 74–80, 2013, doi: 10.1145/2408776.2408794.
- [14] F. Chollet et al., “*Isolation Forest for anomaly detection in large-scale cloud environments,*” IEEE Access, vol. 8, pp. 93215–93228, 2020, doi: 10.1109/ACCESS.2020.2994517.

- [15] F. T. Liu, K. M. Ting, and Z. Zhou, “*Isolation Forest*,” in Proc. IEEE Int. Conf. Data Mining (ICDM), 2008, pp. 413–422, doi: 10.1109/ICDM.2008.17.
- [16] A. Verma, G. Das, P. Dutta, N. Roy, and A. Bhattacharya, “*Anomaly detection in cloud computing systems: A survey*,” Future Generation Computer Systems, vol. 120, pp. 1–18, 2021.
- [17] T. Chen and C. Guestrin, “*XGBoost: A scalable tree boosting system*,” in Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD), 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [18] T. Ouyang et al., “*Training language models to follow instructions with human feedback*,” arXiv preprint arXiv:2203.02155, 2022. Available: <https://arxiv.org/>
- [19] OpenAI, “*GPT-4 Technical Report*,” arXiv preprint arXiv:2303.08774, 2023. Available: <https://arxiv.org/>
- [20] Google DeepMind, “*Gemini: A Family of Highly Capable Multimodal Models*,” arXiv preprint arXiv:2312.11805, 2023. Available: <https://arxiv.org/>
- [21] P. Lewis et al., “*Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*,” Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, pp. 9459–9474, 2020.
- [22] J. Guo, Y. Wang, and X. Li, “*AIOps: Real-time IT operations through machine learning and data analytics*,” IEEE Access, vol. 9, pp. 104313–104329, 2021.
- [23] L. Xu, H. Wang, and Y. Zhang, “*Self-healing cloud infrastructure using intelligent monitoring and automated remediation*,” Future Internet, vol. 14, no. 8, pp. 1–17, 2022.
- [24] C. Delimitrou and C. Kozyrakis, “*Paragon: QoS-aware scheduling for heterogeneous datacenters*,” ACM SIGPLAN Notices, vol. 48, no. 4, pp. 77–88, 2013, doi: 10.1145/2499368.2451155.
- [25] B. Sigelman et al., “*Dapper, a large-scale distributed systems tracing infrastructure*,” Google Research, Tech. Rep., 2010.